

AI-Enabled Learning & Competency Gap Platform

iGOT Karmayogi Native Design Integration, UI Breakdown & Group-Wise Feature Specification

1. Executive Analysis: How Our Platform Integrates with iGOT Karmayogi

Based on the official SIH problem statement and the 18 provided screenshots of the iGOT Karmayogi ecosystem, our platform operates as an **Intelligent Companion Hub** that seamlessly integrates directly into the official portal. In the screenshots (specifically Screen 12 and Screen 15), under *Explore > COMING SOON*, iGOT has explicit placeholders for **"Competencies (Explore your competencies)"** and **"Careers"**. Our platform directly fills this vital missing piece! Rather than replacing iGOT, our portal connects via SSO (Parichay/iGOT IdP) and Karmayogi API adapters to provide continuous, AI-driven skill auditing, customized practice assessments, gap radar analytics, and blockchain-anchored certifications.

2. Deconstructed UI Patterns from iGOT Screenshots (Design System)

To guarantee maximum adoption and evaluator credibility during SIH evaluations, our frontend **MUST** match the exact visual language and UI patterns captured in the screenshots:

Screenshot / Module	Observed UI / UX Pattern	Platform Integration Feature
1. Home & Spotlight (1.jpeg)	Deep Navy Appbar (#08214D), user initials badge (BS), Notification bell with count badge, Pill Search bar, In Spotlight circle icon grid (Bharat Kalp, Marketplace, Specializations, MDO Channel), Banner carousel, Continue Learning card with bottom audio/video bar.	Embed our "AI Competency Diagnostics" banner and spotlight launcher directly into the user Home view.
2. Bharat Kalp & Programs (2.jpeg, 6.jpeg, 7.jpeg)	Multi-week timeline tracker (Week 1..Week 8 in progress), Circular radial progress badges (0%, 100%), Curated Program card with partner logos, star ratings (4.5, 4.6), Provider attribution ("By Karmayogi Bharat").	Feed our AI Dynamic Pathways (Now/Next/Later) matching this exact weekly milestone and circular progress UI.
3. Content Partners (3.jpeg)	Grid cards with rounded borders, circular partner brand avatars (Coursera, Harvard Business, eCornell, NIUA/NULP), "View Provider >" CTA.	External and ministry content catalog filters for statistical modules (MOSPI, ISI Kolkata, World Bank Data).
4. MDO & Channels (4.jpeg, 9.jpeg, 10.jpeg, 18.jpeg)	Color-coded circular avatar chips with ministry/department initials (DO, CT, DP, MO, NE), alphabetical list and A-Z filtering.	Role-based competency profiles customized per Ministry (MOSPI, DES State Directorates, Railways, Agriculture).
5. Explore & Hubs (11.jpeg, 12.jpeg, 15.jpeg, 17.jpeg)	Grouped item cards with icons: Hubs (Discuss, AGK knowledge base, Events), Do More (Learner Passbook, Explore Content, Karma Programs), and Coming Soon list.	Integrate our "GapRadar Analytics" and "Adaptive Assessment Runner" right inside the Competency Hub.
6. Achievements & Karma (13.jpeg, 14.jpeg, 16.jpeg)	4-pack stats cards (Karma Points, Rank, Badges, Learning Hours), Weekly Claps streak timeline (W1..W4 checks), 3D Pyramid Podium Leaderboard for Top Karmayogis, monthly rank delta ("Up by 522 rank").	Award Karma Points for completed competency assessments and link NFT certifications to the official Passbook.
7. Profile & Menu (5.jpeg, 8.jpeg)	Centered user avatar with profile completion progress ring (orange/green circle), email, mobile number, clean single-line menu list with chevrons.	Display official designation, assigned role taxonomy, verified competencies, and Web3 wallet address.

3. Comprehensive Features Required Across Groups A, B, and C

To build a winning prototype that satisfies both the SIH requirements and aligns with the iGOT ecosystem, the following features are allocated across the three specialized development teams:

A. Group A (Frontend Team — M1 & M2): iGOT Native Web/App Portal

Tech Stack: Next.js 14 (App Router), Tailwind CSS, Lucide Icons, Recharts, Axios, i18next (Hindi/English).

- **iGOT Themed Design System:** Pixel-perfect recreation of iGOT color scheme (Deep Blue #08214D, Active Blue #1E50A0, White/Neutral card backgrounds, rounded cards, official badges).
- **Dual Navigation Experience:** Responsive desktop sidebar navigation + mobile bottom-tab bar (Home, Explore, My Learning, Achievement, Competencies, AI Assistant).
- **Live GapRadar Component:** Recharts-powered radar/polar chart showing competency level vs required role baseline across 6-8 statistical domains.
- **Adaptive Assessment Runner:** Single-question stepped view with countdown timer, difficulty indicator, question flag, instant explanation modal, and detailed competency scorecard.
- **Interactive Learning Path (Now / Next / Later):** Kanban/Milestone view of recommended iGOT courses featuring provider tags, match score %, and deep links.
- **Achievement & Karma Center:** Profile stats cards, 4-week engagement streak, pyramid leaderboard for department statisticians, and earned badges.
- **Blockchain Verifiable Passbook:** Certificate showcase displaying Polygon NFT badge, transaction hash, token ID, and verify-on-chain modal.
- **Material Ingestion & Quiz Studio:** Admin/Teacher interface to drag-and-drop statistical manuals (PDF) and immediately review/edit AI-generated MCQs before publishing.
- **Hindi / Regional Language Switcher:** Complete i18n support for government statisticians across state directorates.

B. Group B (Backend & AI/ML Team — M3 & M4): Core Engine & RAG

Tech Stack: Python FastAPI, PostgreSQL (12 Entities), pgvector, Redis, PyJWT, Gemini API / LangChain, LightGBM.

- **FastAPI Microservices:** High-throughput REST API with dependency injection, auto-generated OpenAPI docs, and Redis response caching.
- **12 Core Data Models:** User, Role, Competency, RoleCompetency, Evidence, Assessment, Question, LearningResource, Recommendation, LearningPath, TelemetryEvent, AuditLog.
- **Dynamic Gap Scoring Engine:** Calculates gap vector: $\text{Gap} = \max(0, \text{TargetLevel} - \text{DemonstratedLevel}) * \text{RoleWeight}$ with confidence penalties.
- **Transparent Weighted Recommender:** $\text{Score} = 0.40(\text{Gap}) + 0.20(\text{Level}) + 0.15(\text{Duration}) + 0.10(\text{Evidence}) + 0.10(\text{Language}) + 0.05(\text{Quality})$ with JSON reason codes.
- **Document Ingestion & RAG Pipeline (M4):** Chunking statistical documents (300-800 tokens), generating embeddings, and storing in pgvector with cosine similarity search.
- **Gemini Grounded MCQ Generator (M4):** Strict JSON generation of 4-option MCQs with difficulty, Bloom taxonomy level, distractor rationale, and document citation locator.
- **Mock iGOT Adapter:** Standardized interface simulating Karmayogi course catalog, user enrollments, and completion webhooks.
- **RBAC & Parichay Auth Simulation:** JWT auth with roles: Learner (Statistician), Department Head, Subject Matter Expert, and Platform Admin.

C. Group C (n8n, Blockchain & DevOps Team — M5 & M6): Automation & Trust

Tech Stack: n8n (Self-Hosted), Solidity, Hardhat, Polygon Amoy Testnet, Web3.py, Docker Compose, MiniIO, Grafana.

- **10 Core n8n Workflows (M5):** Automated pipelines for iGOT Catalog Sync (WF-01), Profile Sync (WF-02), Baseline Assessment scoring (WF-03), Async Recommendation generation (WF-04), Document OCR & Ingestion (WF-05), Quiz generation queue (WF-06), Human review routing (WF-07), Telegram/WhatsApp reminders (WF-08), Telemetry normalization (WF-09), and Due Re-assessment trigger (WF-10).

- **Smart Contracts on Polygon Amoy (M6):** *CompetencyCertificate.sol* (ERC-721 soulbound/verifiable NFT) and *AuditAnchor.sol* (tamper-proof event hashes).
- **IPFS / Pinata Storage:** Off-chain decentralized storage for certificate metadata, badges, and assessment provenance records.
- **FastAPI Web3 Microservice:** Python service using Web3.py to sign transactions, estimate gas, and mint certificates automatically upon passing score.
- **Unified Docker Compose Stack:** One-click orchestration for PostgreSQL, Redis, MinIO (S3 compatible), n8n, FastAPI backend, and Next.js frontend.
- **Observability & SIH Evaluation Dashboard:** Prometheus + Grafana dashboard monitoring API latency, n8n workflow execution rates, LLM token burn, and blockchain transaction counts.

4. Complete Website Site-Map & Page Specifications

The frontend website should be structured as follows to match the iGOT screenshots while integrating all AI capabilities:

Route / Page	iGOT Visual Equivalent	Key Features & Interactive Elements
/ (Root) & /login	iGOT Parichay SSO Login screen	Official government portal header, Parichay SSO login button, demo quick-login (Learner, Officer, Admin), Hindi/English language toggle.
/dashboard (Main Hub)	Home Tab (1.jpeg, 11.jpeg)	Spotlight quick-links, Continue Learning resume bar, GapRadar interactive chart, Quick assessment CTA, Karma Points & Weekly Clap streak widget.
/competencies (Gap Engine)	Explore > Competencies (12.jpeg Coming Soon)	Detailed breakdown of official statistical competencies (Sampling, Survey Design, Regression, National Accounts). Required vs actual sliders and priority score badges.
/assessment (Runner)	iGOT Assessment Module	Clean assessment runner: timed questions, MCQ options, difficulty badge, provenance source badge ("Extracted from MoSPI Manual 2024"), instant results screen.
/learning-path (Recommendations)	My Learning & Bharat Kalp (2.jpeg, 6.jpeg, 11.jpeg)	Personalized courses ranked with Explainable AI badges ("40% Gap Match • High Urgency"). Tabs: Now / Next / Later. Filter by Provider (Coursera, eCornell, MoSPI).
/achievements (Certificates)	Achievement Tab (13.jpeg, 14.jpeg, 16.jpeg)	Leaderboard pyramid for department statisticians, Karma point history, Polygon Blockchain NFT Certificate gallery with "Verify on PolygonScan" links.
/documents (AI Studio)	Admin / MDO Knowledge Hub (17.jpeg AGK)	Upload statistical manuals & survey reports (PDF). Live chunking & embedding progress bar, generated question preview with SME approve/reject buttons.
/analytics (Department)	MDO Channel / Admin View (4.jpeg, 18.jpeg)	Heatmap of departmental competency readiness, completion rates across ministries/states, exportable PDF reports for leadership.

5. Definitive Answer: Does Our Website Integrate into iGOT?

YES. That is the exact premise of this SIH submission. As shown in Screenshot 12 & 15, iGOT already has a placeholder for "Competencies — Explore your competencies (Coming Soon)". By building this platform with the exact visual language, SSO hooks, and iGOT catalog integration:

- 1. Zero User Friction:** Government statisticians access our tools directly within their existing learning routine without learning a new, disconnected portal.
- 2. Instant Evaluation Impact:** SIH evaluators from Karmayogi Bharat and MoSPI will immediately see this as a production-ready module that can be directly deployed into iGOT.
- 3. Pluggable Micro-Frontend Architecture:** The website can run both standalone (for departmental analytics) and as an embedded iFrame / Web Component inside the iGOT Mobile & Web portal.

6. Immediate Next Steps for the Team

- **Group A:** Implement the frontend theme matching iGOT Navy (#08214D) and build the 7 key routes.
- **Group B:** Stand up the FastAPI backend with the 12 models, gap scoring formula, and Gemini RAG endpoint.
- **Group C:** Spin up Docker Compose (Postgres, Redis, MinIO, n8n) and deploy the ERC-721 contract on Amoy.